Exp : 5 VPS / Virtual Machine Setup and Linux Configuration

Aim

To provision a Virtual Private Server (VPS) or create a virtual machine using virtualization software such as **VirtualBox** or **VMware Workstation**, install a Linux operating system, and configure network settings including IP address, subnet mask, gateway, and DNS.

Theory

A **Virtual Private Server (VPS)** is a virtualized environment that acts like a dedicated server. It can be hosted on cloud platforms or created locally using virtualization tools. Virtualization software such as:

- VirtualBox
- VMware Workstation allows running multiple operating systems on a

single physical machine. Linux distributions like:

- Ubuntu
- CentOS
- Debian

are commonly used due to stability, security, and open-source nature.

Procedure

Step 1: Install Virtualization Software
Option A: Install VirtualBox

1. Download from official website
2. Install with default settings

Option B: Install VMware Workstation

1. Download installer
2. Follow installation wizard

Step 2: Create Virtual Machine In VirtualBox:

1. Open VirtualBox → Click **New** 2. Enter:
    - Name: LinuxVM
    - Type: Linux
    - Version: Ubuntu (64-bit)
3. Allocate:
    - RAM: 2 GB (minimum)
    - CPU: 1–2 cores
4. Create Virtual Hard Disk:
    - Type: VDI
    - Size: 20 GB

Step 3: Attach ISO File

1. Download ISO (e.g., Ubuntu)
2. Go to VM Settings → Storage
3. Attach ISO file

Step 4: Install Linux OS

1. Start VM
2. Select **Install Ubuntu**
3. Follow steps:
    - Language selection
    - Keyboard layout
    - Installation type (Normal)
    - Create username & password
4. Click **Install**
5. Restart VM after installation

Step 5: Configure Network (Automatic - DHCP)
Check IP: ip a
Check connectivity: ping
google.com

Step 6: Configure Static IP Address Edit
Netplan config:
sudo nano /etc/netplan/01-netcfg.yaml
Example configuration: network:
version: 2 ethernets: enp0s3:
dhcp4: no addresses:
[192.168.1.100/24] gateway4:
192.168.1.1 nameservers:
addresses: [8.8.8.8,
8.8.4.4] Apply changes: sudo
netplan apply

Step 7: Verify Network Configuration
Check IP: ip a
Check gateway:
ip route Test
DNS:
ping google.com

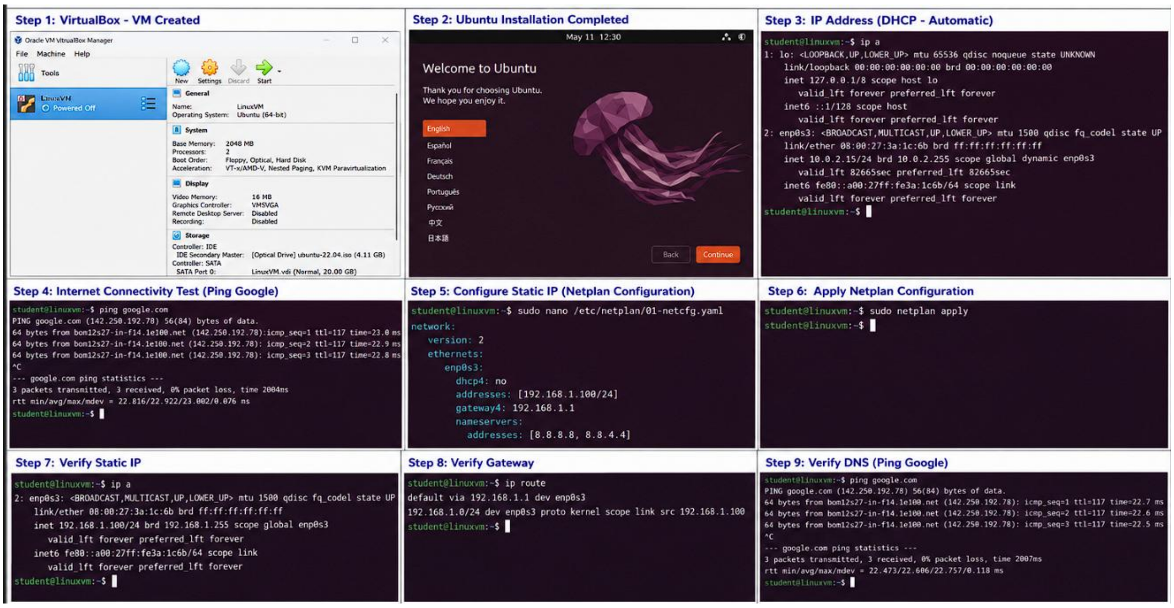
Step 8: (Optional) VPS on Cloud Instead
of local VM, you can use:

- Microsoft Azure
- Amazon EC2

Steps:

1. Create VM instance
2. Select Linux image
3. Configure networking
4. Connect via SSH

OUTPUT:



Result

Successfully created a virtual machine, installed Linux OS, and configured network settings including IP address, gateway, and DNS.